Al-Balqa' Applied University
Faculty of Engineering Technology, Department of Basic Scientific Sciences
Final Exam . Linear Algebra I

Date: 14/ 6/2022

Part A: Multiple choice (24 marks)

-Be sure that you have 12 different questions.
-Complete the following table by putting the correct answer letter in the table.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12

Q1) Suppose that the augmented matrix for a system of linear equations has been reduced by row operations to the given form $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$ The solution of the linear system (respectively) x,y,z is

a) 2,-3,2 b) 2, -1, 1 c) -1,5,-1 d) 4, -3, 2

Q2) The values of k, if any that satisfy the equation $\begin{bmatrix} k & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix}\begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$ is a) 2 b) 5 c) 1 d) -1

Q3) If $\left(I - A^T\right)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{-1}{3} \end{bmatrix}$, then $Tr(A) =$ a) 4 b) 3 c) -40 d) 2

Q4) For some nonsingular square matrix A, assume that $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 2 \end{bmatrix}$ the solution of the system $AX = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ is a) $X = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix}$ b) $X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ c) $X = \begin{bmatrix} -1 \\ 1 \\ -4 \end{bmatrix}$ d) $X = \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$

Q5) The system $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

a) no solution b) two solution c) infinitely many solutions d) unique solution

Q6) The values of k for which $\det(A)=0$, where $A = \begin{bmatrix} 7 & 0 & 0 \\ -3 & k-6 & 0 \\ 0 & 0 & k-4 \end{bmatrix}$ is a) 4, 3 b) 1, 6 c) 4, 6 d) 1,-3

Q7) If $A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 4$ evaluate the $\begin{vmatrix} a & b & c \\ -2g & -2h & -2i \\ d & e & f \end{vmatrix} =$ a) -8 b) 12 c) -12 d) 8

Q8) The value of x, if any that satisfy the equation $\begin{vmatrix} 2 & x+2 \\ -1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -9 & x-2 & 0 \\ 8 & 5 & 2 \end{vmatrix}$ is, a) 12 b) 4 c) 2 d) 6

Q9) Let the vectors $v = (2, 3)$, $w = (-4, 8)$ the components of the $v-2w$ is
a) (6, 10) b) (16, 10) c) (6, -10) d) (-6, 17)

Q10) The terminal point of the vector that is equivalent to $u = (2,4,6)$ and whose initial point is B (-3,-1,2),
a) (-1,3,8) b) (8,-1,1) c) (2,-1,8) d) (-1,3,4)

Q11) Let $u=(2,3,0)$ and $v=(-1,2,2)$ the distance between u and v is a) $\sqrt{17}$ b) $\sqrt{14}$ c) $\sqrt{11}$ d) $\sqrt{10}$

Q12) Let $A = \begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ the Eigen values = a) 2, 3 b) -2, 3 c) -2, -3 d) 2, -3

Part B: Fill the blank (12 marks)

Q1: Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ Then $|(2A)^{-1}| =$ _____

Q2: Let $A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ the adj $A^{-1} =$ _____

Q3: Let $V = \mathbb{R}^2$ define the operations $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + 2x_2, y_1 - 3)$ and $c \odot (x, y) = (cx, c + y)$ then $4 \odot ((2,3) \oplus (-2, 5)) =$ _____

Q4: Let $u = (0, 1, -1)$, $v = (1, 2, 0)$ then the cosine of the angle θ between u and v is _____

Q5: Let the vectors $u = (2, k)$ and $v = (3, 6)$, the value of k such that u and v are orthogonal _____

Part C: Written questions (12 marks)

Question 1: Let $A = \begin{bmatrix} 1 & 5 & 6 & 2 \\ 2 & 4 & -2 & 1 \\ 6 & 12 & -6 & 3 \\ 1 & 0 & -1 & 2 \end{bmatrix}$ show that det (A)=0, without directly evaluating the determinant, by state the property which use.

Question 2: let $W = \{f \in p_2 : a_0 = 0\}$ where p_2 set of vectors for 2nd order polynomials. Show that W is a subspace of the vector space.

Question 3: Let $p_1 = 1 + x$, $p_2 = 3 + 3x - 2x^2$, $p_3 = 1 + 2x - x^2$ are p_1, p_2, p_3 linearly dependent or independent? Why.

Question 4: Let det (A)=2, det (B) = 4 find $\det(A^{-1}B)$.

With my best wishes .

$\frac{d}{dt}A^{-1} = \frac{A}{dt}$

$A \times dA^{-1} =$

A^{-1} :

A^{+}

$A = \begin{bmatrix} 3 & & \\ & 3 & \\ & & 3 \end{bmatrix}$

$A^{1} = \begin{bmatrix} \frac{1}{3} & & \\ & \frac{1}{3} & \\ & & \frac{1}{3} \end{bmatrix}$

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$\begin{bmatrix} -2 & -2 \\ -8 & 2 \end{bmatrix}$

$(-8, 5, 2)$

$K = -1$